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EEE 202 Lab 7: Design Project

Data Sheet

Part 1: Voltage Divider Design

a) What do you think the purpose of the diode rectifier is in this design? What voltage drop are you assuming across the diode (V_D)? It directs the current in one direction and makes sure it doesn't flow back into the solar panel

b) Connect RL = 1 kΩ to the solar panel and measure <u>voltage</u> (parallel) and <u>current</u> (series) under the listed light conditions. Use the portable DMM outdoors and the lab multimeter inside. Record all measurements below.

(If there are limitations due to weather or light conditions, talk to your TA about which to complete)

Note the number of the solar panel you are using and use the same one each session: 12

Light Condition	Solar Panel Voltage (V)	Solar Panel Current (mA)
Full Sunlight (outside)	6.83	6.76
Shaded Sunlight (outside)	1.3	1.7
Lab Light (inside)	0.190	0.191
Other (note details):		

c) The CEO of Sparky Solar, Melanie Waters, has told you that the design criteria (in order of importance) are: <u>1. Availability, 2. Cost, 3. Smaller physical size</u>. When designing the voltage divider circuit, which set of resistors will you use (1, 2, or 3) and why? We're going to use the second set of resistors because of the smaller price + availability. The higher max operating temperature also allows this to be used outside as temperature increases

d) Select R1 and R2 to ensure that the value of A1 does not exceed 5V for the maximum solar panel voltage (show your work here): Which light condition should be considered and why?

$V_{A1} = V_1 \left(\frac{R_2}{R_1 + R_2}\right)$

$5V \geq 6.83V \left(\frac{100K}{47k + 100K}\right)$

$5V \geq 4.646V$

Full sunlight should be considered as that is the maximum solar panel voltage

R1 (Ohms)	R2 (Ohms)	Max Voltage in to Pin A1 (V)
470	100	4.

e) What is the <u>maximum voltage</u> across each resistor without exceeding the power rating of your selected R1 and R2? The resistors are ¼ W resistors. ***(If your maximum for either is less than 5V, you will need to reselect your R1 & R2 values).***

$V_{max,R2} = \sqrt{0.25(100)} = 15.81$

$V_{max,R1} = \sqrt{0.25(470)} = 10.83$

Max V_R (V

For R1:

10.83

For R2:

15.81

f) Create a simulation of your Voltage Divider circuit in LTSPICE or TinkerCAD and attach an image of your schematic. Label it "Schematic 1, Part 1".

g) Physically build and test your circuit. Do your measured results from your built circuit match your simulated results? How much error are you seeing? What are some possible reasons for the difference in the real and theoretical results? Our physical build had a result of 4.3V while the simulated was 4.23V, less than 2% error. Some difference was most likely due to resistance in the wires, age & quality of materials, and possibly the breadboard used

Part 2: Op-Amp Design

****** Remember, when measuring current the portable multimeters have current limits (check the multimeter). If your calculations show you will be above the limit of the current input, use the 10A terminal!!! Failure to use the correct terminal will result in blowing the fuses of the multimeter. ******

a) Why do you think the current measurement needs to be done using an op-amp rather than how we measured the voltage in Part 1? The op amp amplifies the very small voltage from the shunt resistor to a larger value (0-5V range) so the Arduino can read it correctly.

hat is the <u>maximum current</u> that the solar panel can output without exceeding the power rating load resistor RL. The resistors provided are ¼ W resistors. *(If your solar panel can exceed this nt value, you will need to a load resistor with a higher power rating).*

$P = I^2R \Rightarrow I_{max} = \sqrt{\frac{P}{R}}$

$= \sqrt{\frac{0.25}{1000}} = 0.0158A$

$V^+_{max} = I_{max} \cdot R_{sh} = 0.0158(50) = 0.791V$

) Using the <u>highest</u> load current, determine the maximum voltage into the + terminal of the op-amp:

Max I_{load} (mA)	R_{sh} (Ohms)	Max V^+ (mV)
15.81	50	791

d) The CEO of Sparky Solar, Melanie Waters, has told you that the design criteria (in order of importance) are: <u>1. Higher Temp, 2. Sooner availability, 3. Lower cost</u>. When designing the op-amp circuit, which set of resistors will you use (1, 2, or 3) and why? we chose set 2 as it had one of two higher max operating temps, and readily available.

e) Select Ri and Rf to boost the output voltage, but ensure that the Op-Amp output voltage does not exceed 5V for all light conditions (show your work here):

$A_v = \frac{V_{out}}{V^+} = \frac{4.1895}{0.790} = 5.24$

$A_v = 1 + \frac{R_f}{R_i}$

$1 + \frac{200(10^3)}{22(10^3)}$

$= 5.24$

$Gain = \frac{4.301}{0.79} = 5.546$

$V_{out} = \left(1 + \frac{100(10^3)}{22(10^3)}\right)(0.79)$

$= 4.381V$

Ri (Ohms)	Rf (Ohms)	Gain (V/V)	Max Vo (V)
22k	100k	5.5	4.301

f) Why is a rail-to-rail op-amp required for this design? A rail-to-rail op-amp costs more than a standard op-amp, what value to the stakeholder does this extra feature add to the design?

You need rail to rail because it allows full swing to the power rails from 0 to 5 volts which is needed for these parts. Standard/normal op-amps don't allow for that output voltage range

g) Do the values of Ri and Rf need to be extremely accurate for your current measurement to be precise? Explain why or why not.

No they do not. They simply need to have a max output voltage as close to 5V while using the required resistors

h) Create a simulation of your Op-Amp circuit in LTSPICE or TinkerCAD and attach an image of your schematic. Label it "Schematic 1, Part 2".

i) Physically build and test your design. Do your measured results from your built circuit match your simulated results? How much error are you seeing? What are some possible reasons for the difference in the real and theoretical results? The measured results are similar to the simulated results. Possible reasons for the difference may be due to the resistance tolerances in the resistor of poribble faulty equipment

Part 3: Controller Circuit Design

a) Explain the Voltage, Current, and Power measurement calculations in the provided Arduino code. Include the equations you are using for each.

Equation for Serial Port Display Voltage (V): $V = V_{temp} \cdot \left(\frac{5.0}{1024.0}\right) \cdot \left(\frac{R_1+R_2}{R_2}\right)$

$V_{temp} \cdot \frac{5.0}{1024.0}$: converts digital bit value into actual voltage for arduino to read

$\frac{R_1+R_2}{R_2}$: scales measured pwr voltage back up to solar panel voltage

Equation for Serial Port Display Current (mA): $I_{mA} = \left(\frac{I_{temp} \cdot \left(\frac{5.0}{1024.0}\right)}{Gain \cdot R_{SH}}\right) \cdot 1000$

$I_{temp} \cdot \frac{5.0}{1024.0}$: voltage output from op-amp at pin A0

$Gain \cdot R_{SH}$: op-amp multiplied shunt voltage by gain, divided by both to isolate I_{mA}

* 1000 : Amperes to milliamperes

Equation for Serial Port Display Power (mW): $P_{mW} = V_{power} \times I_{mA}$

Formula for power calculation

b) Record the values you are using for: R1 = 47K , R2 = 100K , Ri = 22K , Rf = 100K

c) Calculate the expected serial port results for the following:

A1 (V)	Calculated Voltage Value (V)	A0 (V)	Calculated Current Value (mA)	Calculated Power Value (mW)
0.5	0.735	0.5	1.60	1.72
1	1.470	1	3.61	5.71
2	>940	2	7.21	21.20

d) Create a simulation of your Controller circuit in TinkerCAD and attach an image of your schematic. Label it "Schematic 1, Part 3".

e) Simulation Results: (readings from the simulated Serial Monitor in TinkerCAD)

A0 (V)	A1 (V)	Serial Port Voltage (V)	Serial Port Current (mA)	Serial Port Power (mW)
1	1	1.47	3.61	5.31
3	3	4.41	10.81	47.65
5	5	7.34	14.02	172.28

f) Physical Results: (readings from the physical Serial Monitor in Arduino IDE)

A0 (V)	A1 (V)	Serial Port Voltage (V)	Serial Port Current (mA)	Serial Port Power (mW)
1	1	1.51	3.5	5.28
3	3	4.34	10.57	45.88
5	5	7.24	17.3	126.12

g) Create a simulation of your Voltage Divider and Controller circuit in TinkerCAD and attach an image of your schematic. Label it "Schematic 2, Part 3".

h) Simulation Results: (readings from the simulated Serial Monitor in TinkerCAD)

V_{solar} max (V)	Serial Monitor Voltage Reading (V)
6.83	6.5

i) Do your simulated results from your TinkerCAD circuit match your theoretical results? How much error are you seeing? What are some possible reasons for the difference? The simulated results match closely to the theoretical results, with possible differences are the tolerances in resistance and also the diode.

j) Physical Results: (readings from the physical Serial Monitor in Arduino IDE)

V_{solar} max (V)	Serial Monitor Voltage Reading (V)
6.83	6.6

k) Do your measured results from your built circuit match your simulated results? How much erro are you seeing? What are some possible reasons for the difference in the real and theoretical results

The measured results are slightly off, most likely due to slight variation in resistor values

l) Create a simulation of your Voltage Divider and Op-Amp Circuit and Controller circuit in TinkerCAD and attach an image of your schematic. Label it "Schematic 3, Part 3".

m) Simulation Results: (readings from the simulated Serial Monitor in TinkerCAD)

Vsolar max (V)	Load Current max (mA)	Serial Port Voltage (V)	Serial Port Current (mA)	Serial Port Power (mW)
6.83	6.5	6.6	18.02	118.33

n) Do your simulated results from your TinkerCAD circuit match your theoretical results? How much error are you seeing? What are some possible reasons for the difference?

The simulated results matched very closely with theoretical results, having little error

o) Physical Results of Volt/Amp/Wattmeter: (*do this outside with actual sunlight or the equivalent*)

Vsolar max (V)	Load Current max (mA)	Serial Port Voltage (V)	Serial Port Current (mA)	Serial Port Power (mW)
6.83	6.064	6.51	17.01	110.86

p) Do your measured results from your built circuit match your simulated results? How much error are you seeing? What are some possible reasons for the difference in the real and theoretical results?

The measured results were very slightly off from the simulated results, most likely due to slight difference in resistor values & wire internal resistance

q) What adjustments can you make to the code to increase the accuracy of the voltage, current, and power that display in the serial port?

In order to increase the accuracy we can change the output of the serial print from its default of 2 decimals to 5.

(BONUS POINTS) Note and detail any additional automation you added on a separate page. Include your updated code and schematic attached to the Data Sheet. Bonus points will be given for successful changes in the code, and/or successful changes in the simulation, and/or successful changes in the physical prototype. (TA needs to sign off on successful changes in the simulation or prototype here:

_____(TA initials) Bonus Points awarded: _____pts

TA can assign extra credit points (up to 2pts) to reward teams that have added noteworthy automation to their designs.

Part 4: Complete Automated Solar Panel Power Meter Design

a) Create a simulation of your Automated Solar Panel Power Meter circuit in TinkerCAD and attach an image of your schematic. Label it "Schematic 1, Part 4".

b) Simulation Results: (TinkerCAD)

Solar Panel Voltage (V)	Solar Panel Current (mA)	LCD Voltage Readout (V)	LCD Current Readout (mA)	LCD Power Readout (mW)
2	1.40	2.00	1.99	2.78
4	3.30	4.00	7.28	13.10
8	7.12	6.29	7.04	51.85
10	9.03	7.74	8.95	69.65

c) Did you have to make any component or code changes? If so, what changes were made and why?

Only the addition of the LCD screen changed the circuit. The code changed as the LCD code needed to be implemented into the original code for the circuit.

d) Physical Results: (Build circuit with Arduino board)

Solar Panel Voltage (V)	Solar Panel Current (mA)	LCD Voltage Readout (V)	LCD Current Readout (mA)	LCD Power Readout (mW)
Full Sun = 6.83	5.00	6.56	5.44	36.1
Shaded = 1.30	0.5	1.27	0.63	0.74
Inside Light Source = 0.18	0.00	0.17	0.00	0.00

e) Do your measured results from your built circuit match your simulated results? How much error are you seeing? What are some possible reasons for the difference in the real and theoretical results?

Yes, the measured results match the simulated

(BONUS POINTS) Note and detail any additional LCD functions you added on a separate page. Include your updated code and schematic attached to the Data Sheet. Bonus points will be given for successful changes in the code, and/or successful changes in the simulation, and/or successful changes in the physical prototype. (TA needs to sign off on successful changes in the simulation or prototype here:

_____(TA initials) Bonus Points awarded: _____pts

TA can assign extra credit points (up to 2pts) to reward teams that have added noteworthy LCD functions to their designs.

Part 5: Load Resistor Discharge Rate

Using the First Order Differential, First Order Step-by-Step method, or Laplace transformations, determine the size of the load capacitor to meet the design specifications - determine an appropriate capacitor value (from your available parts) to ensure the load will discharge to 1mV in about 10 msec with a solar panel supply of 6V and 5mA. You will still assume the load resistance (RL) of 1 kΩ and a shunt resistor of 50Ω. (Show your work here).

$$C\frac{dV_c}{dt} = -\frac{V}{R_{eq}} \qquad R_{eq} = R_L + R_{sh} = 1050\,\Omega$$

$$\frac{dV_c}{dt} + \frac{V_c}{cR_{eq}} = 0$$

$$\int \frac{1}{V_c}\,dV_c = \int \frac{-1}{CR_{eq}}\,dt$$

$$\ln(V_c) = \frac{-t}{cR_{eq}} + k$$

$$V_c = V_i e^{-t/cR_{eq}}$$

$$C = \frac{-0.01}{R_{eq}\ln\left(\frac{0.001}{6}\right)}$$

$$= \frac{0.01}{1050(-8.7)}$$

$$C = 1.1\,\mu F$$

(BONUS POINTS) Test your capacitor selection using your physical circuit from Part 4. Measure the discharge time using an oscilloscope and attach an image of your results. Label it "Results, Part 5". (TA needs to sign off on the measurement)

_____(TA initials) Bonus Points awarded: _____pts

TA can assign extra credit (up to 2pts) to reward teams that have built and measured the discharge time.

Prototype Demonstration

Once your physical design is tested in the lab, upload your controller code to the Arduino and rem
the USB cable from the Arduino. Replace the Controller power supply (USB) with the 9V battery -
the power meter should now be portable. Demonstrate your working prototype to the Sparky Solar
Representative (TA) (*in the sun or inside, depending on the conditions*) and get a signature to verify
works correctly.

TA Signature: Liam Smith **Date:** 5/1/26

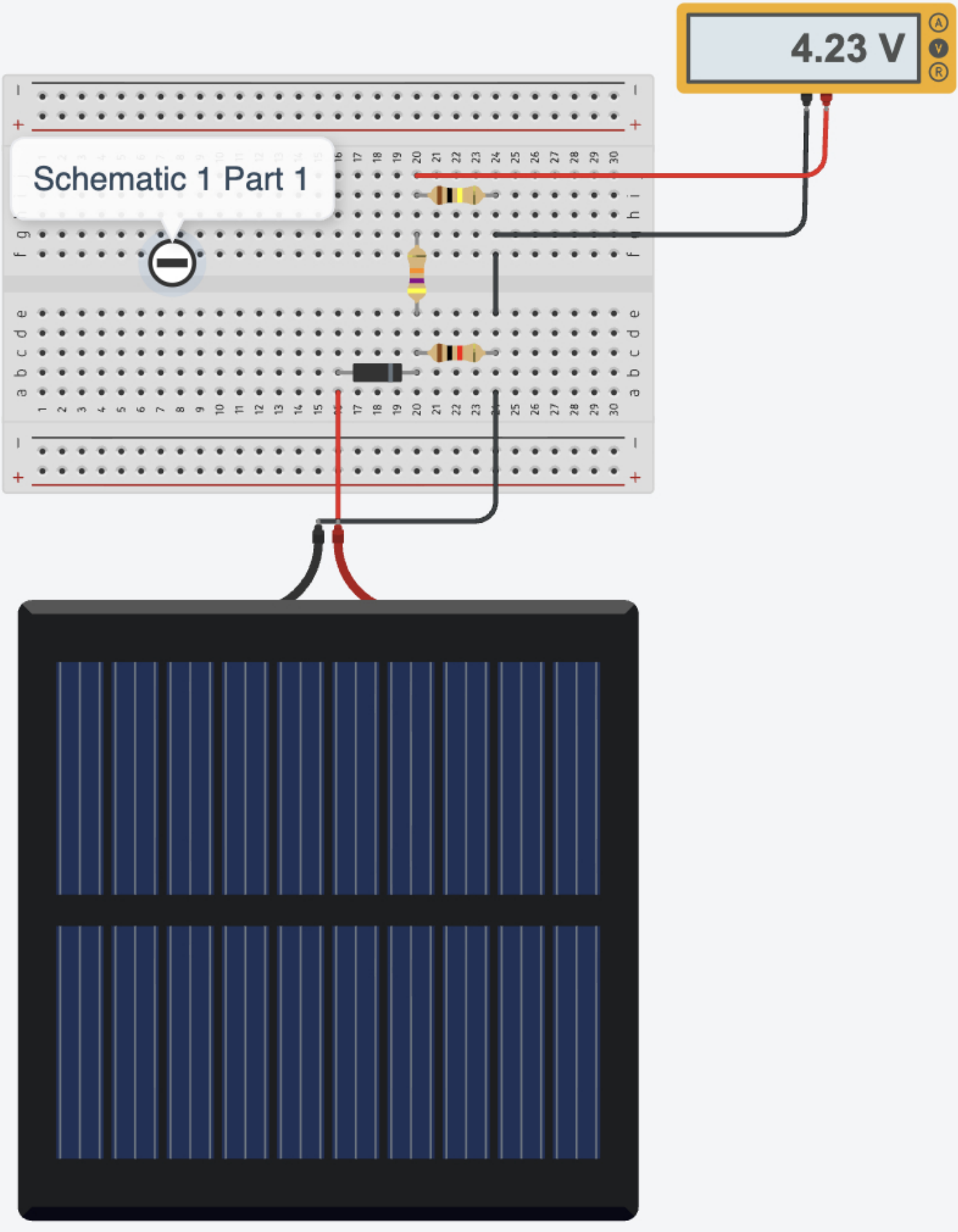
BONUS POINTS – TA to note any additional technical functions, any additional customer considerations, anything that went above and beyond what was asked for.

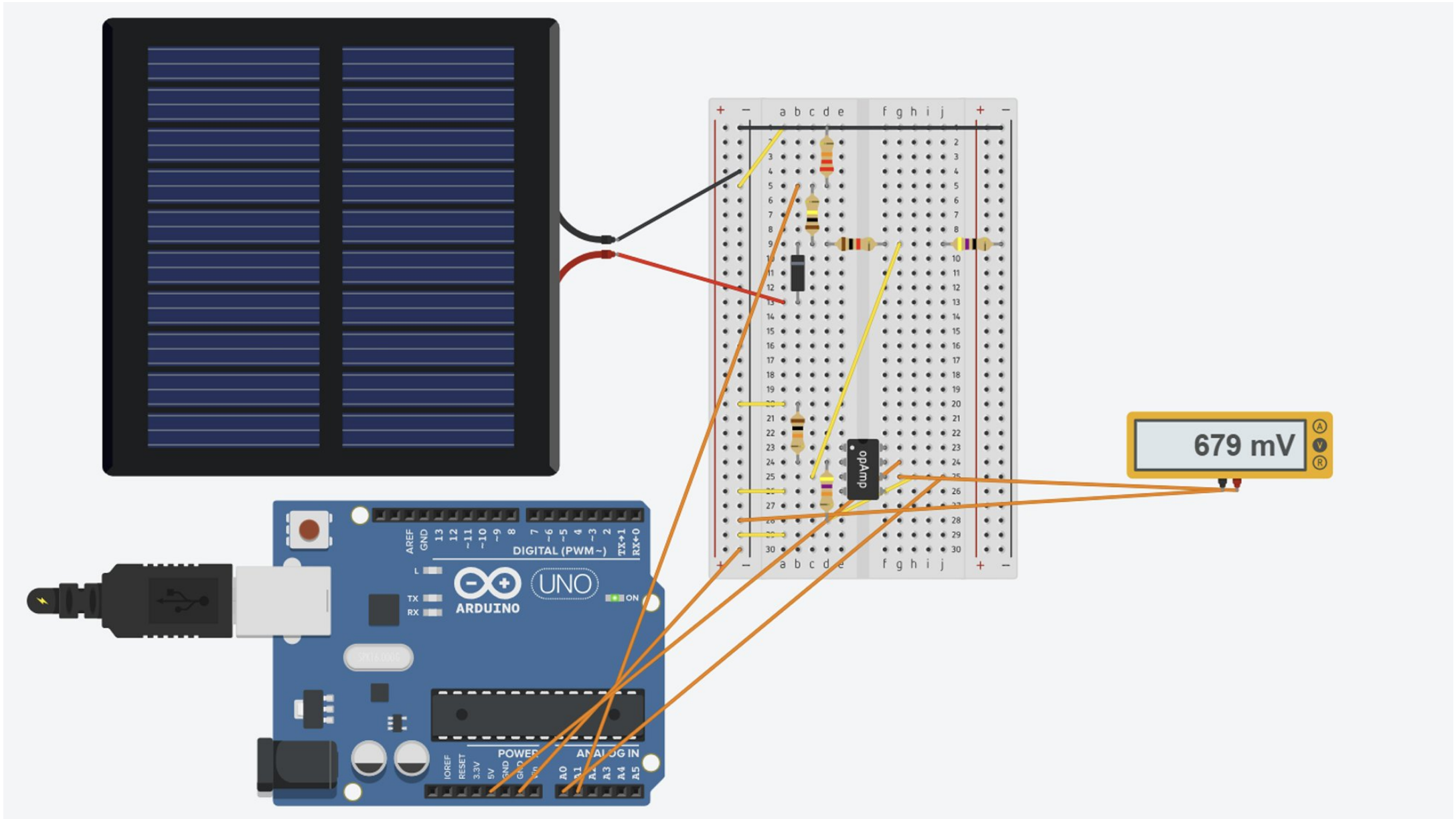
TA can assign extra credit points (up to 4pts) to reward teams that have added noteworthy technical features or demonstrated an entrepreneurial mindset in their final prototype.

TA Notes:

_____(TA initials) Bonus Points awarded: _____pts

Notes: All Lab Data Sheets must be signed by a TA for credit. Failure to get a signature or forging a signature will result in a zero for the lab. You can get a signature late via email with your TA if you forget to do it in the lab. Maintaining a clean and organized lab space is a skill you are also expected to learn. Failure to clean up your space or put components where they belong could result in a 20% penalty at the discretion of the Lab TA.


4.23 V
Schematic 1 Part 1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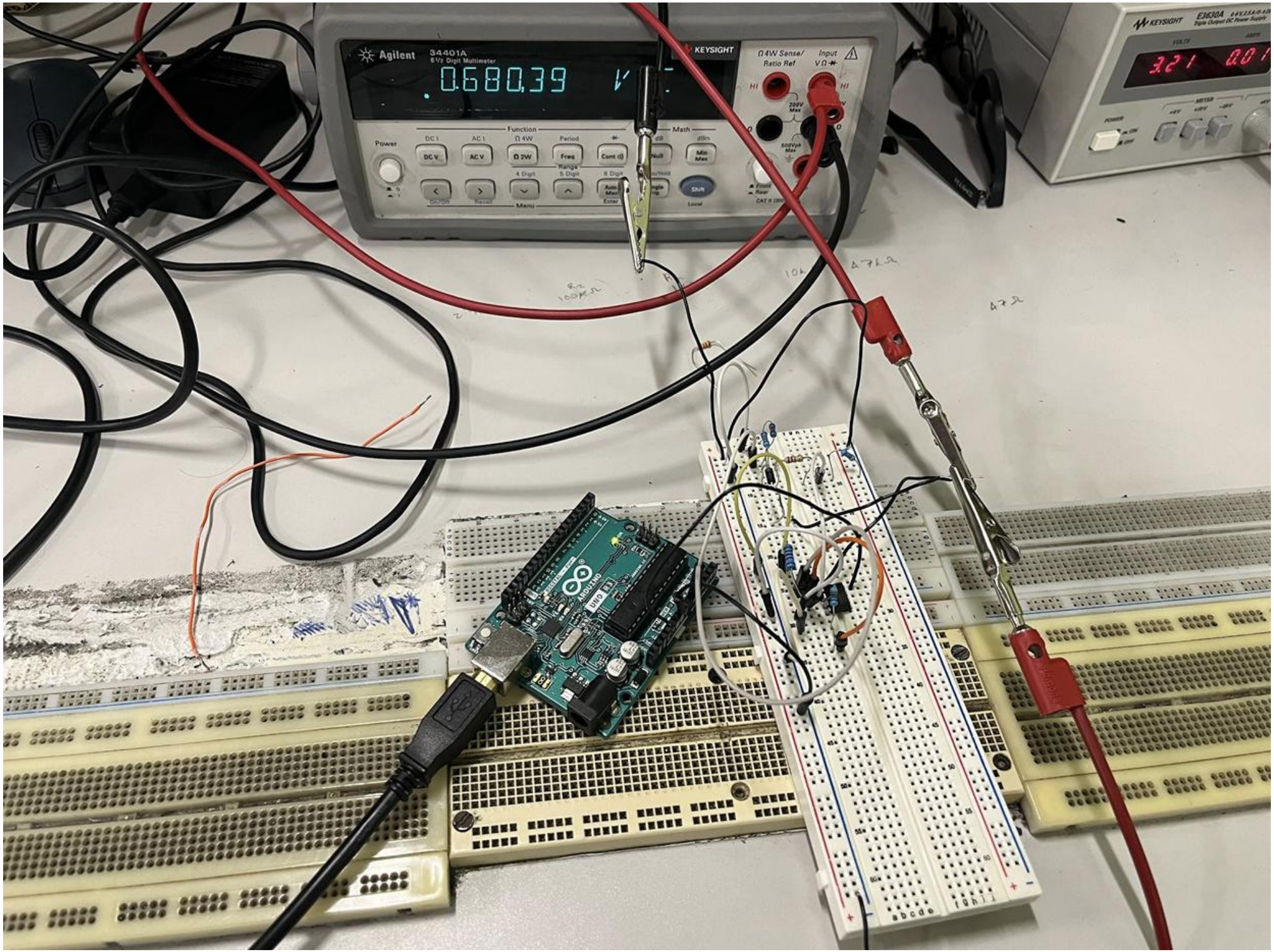
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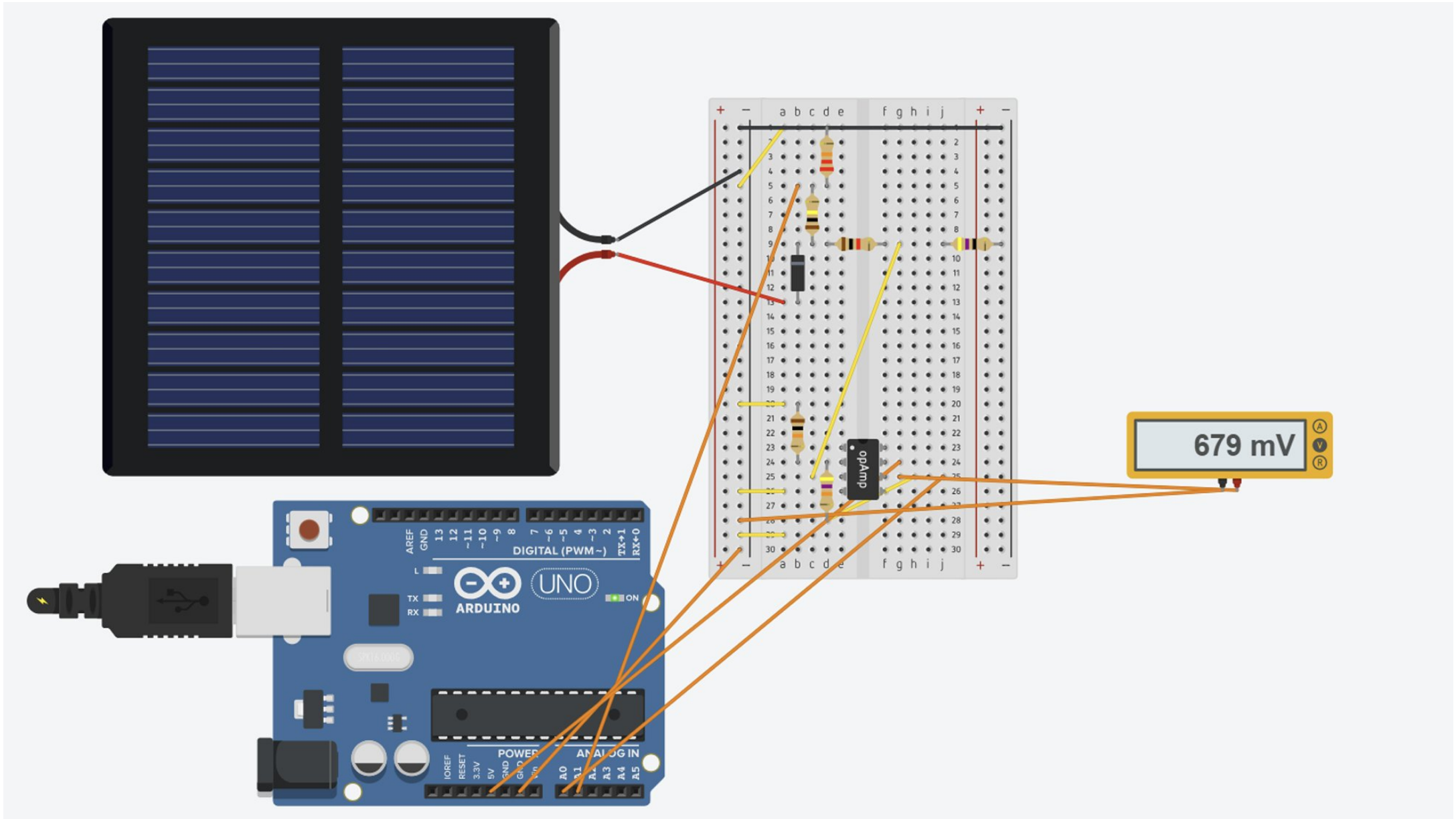
679 mV
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ARDUINO
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POWER
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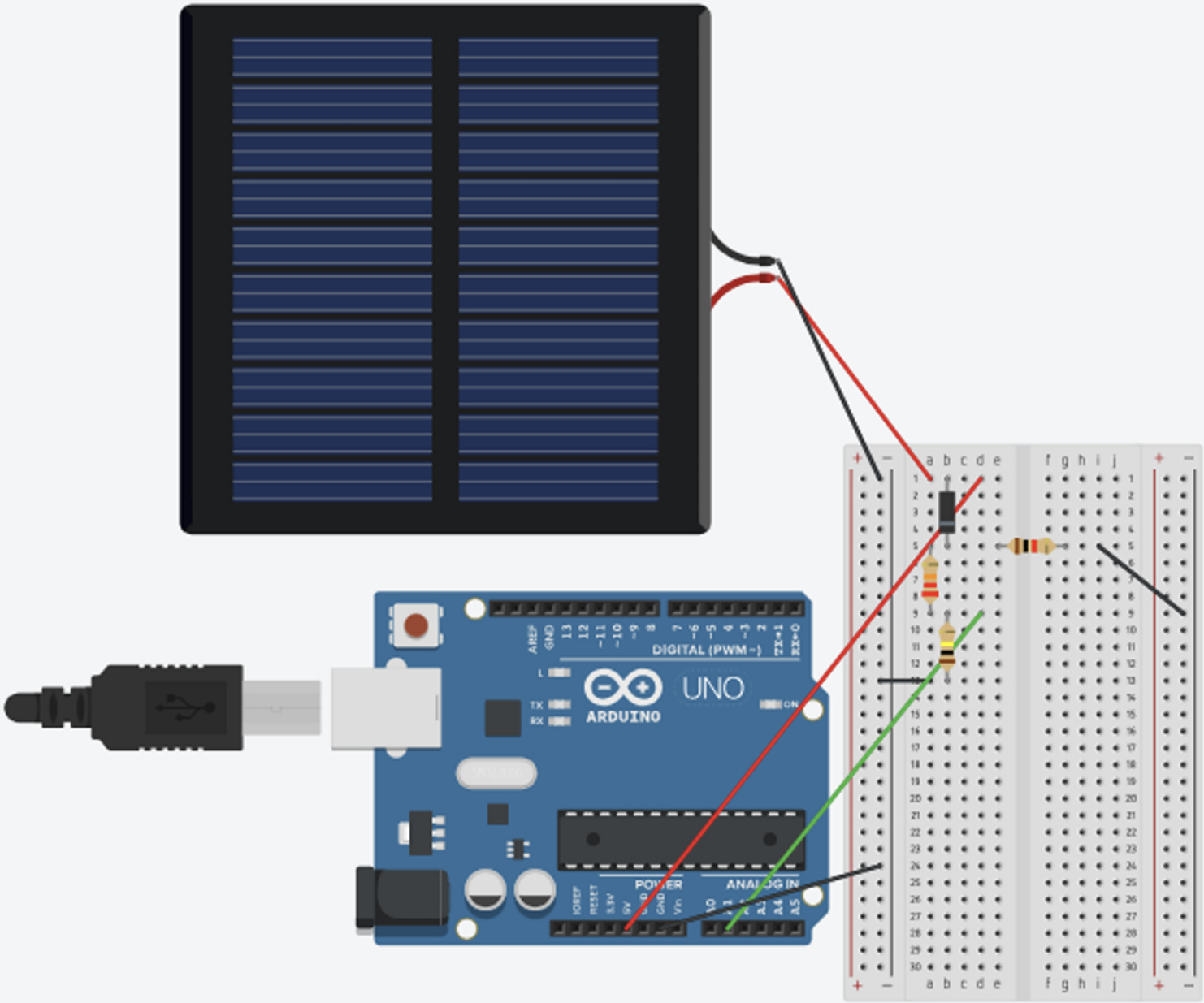
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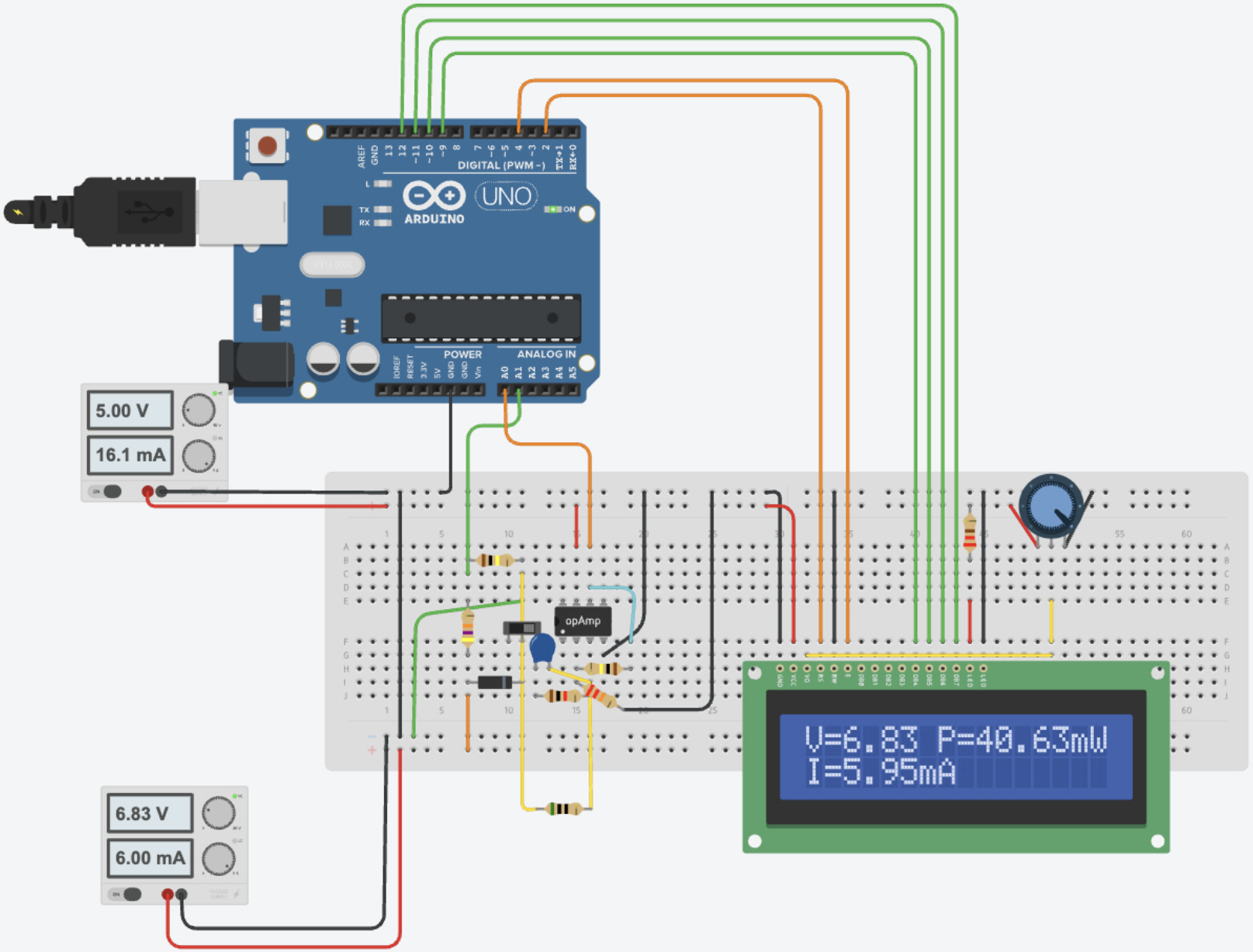
ARDUINO
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5.00 V
16.1 mA
6.83 V
6.00 mA
opAmp
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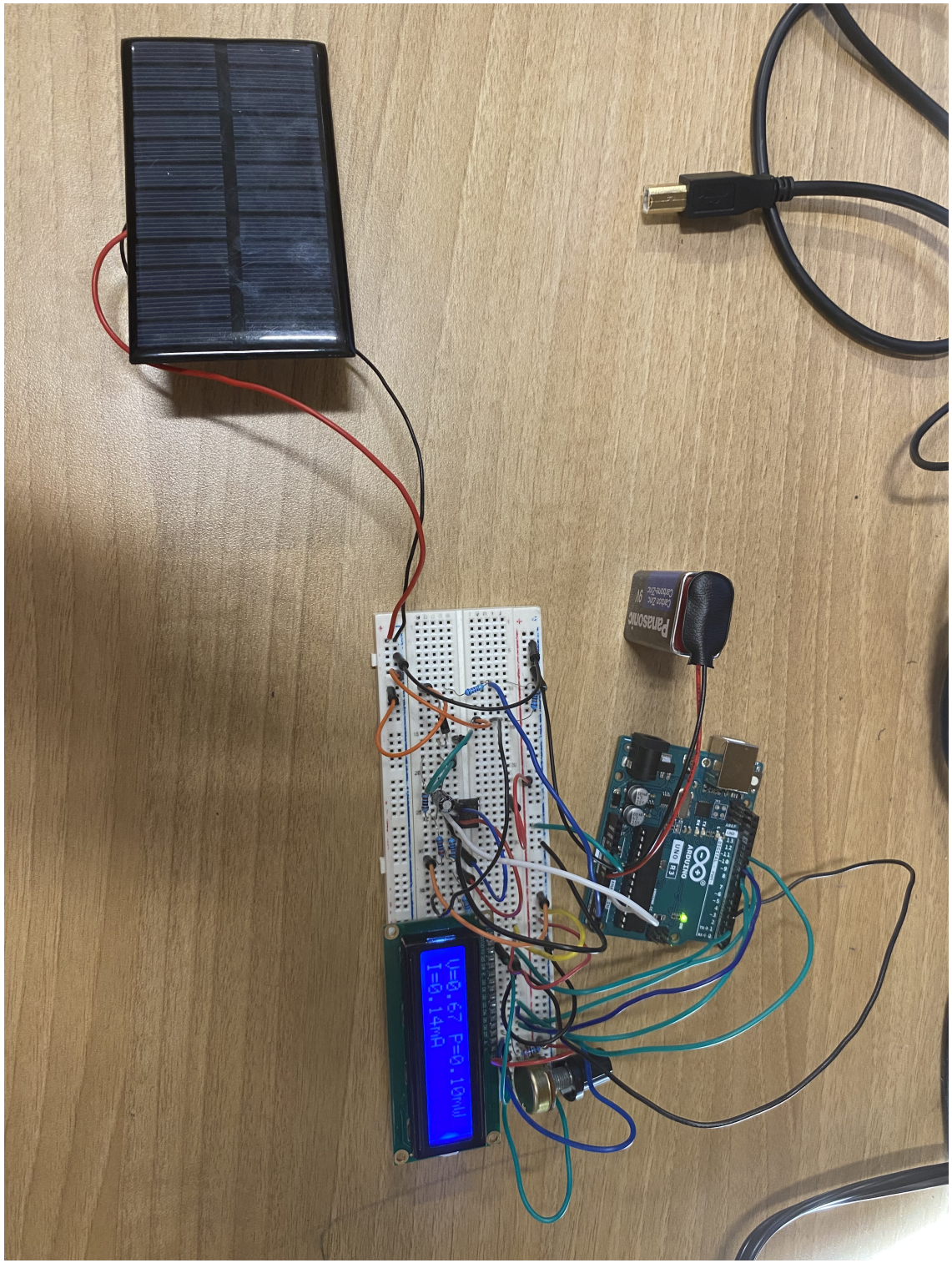
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